

Task 17 — Customer Segmentation (RFM)

Data Analyst · Phase 1 Industry Immersion · Task Brief

DATA ANALYST

PlaceMux · Phase 1 · Task Brief

DIFFICULTY

Mastery · Expert

Level 17 of 20 · ramp 81%

Objective

What this task is: Segment customers using RFM (Recency, Frequency, Monetary) to target and prioritise them by value.

Why it matters: Not all users are equal. RFM turns one undifferentiated base into actionable segments for retention and growth.

What you will deliver

- An RFM segmentation with named segments, sizes, value and recommended actions.

Inputs & prerequisites

- A reproducible Python data environment (pandas, numpy, matplotlib).
- Comfort reading data into a DataFrame and writing basic SQL.
- The habit of validating data before analysing it.
- Required tools for this task: pandas (qcut/rank), seaborn, a segment summary table

Step-by-step execution

Work through these in order. Don't skip the verification steps — that's where 'looks done' becomes 'is done'.

1. Compute recency, frequency and monetary values per customer.
2. Score each dimension into tiers (e.g. quintiles).
3. Combine scores into named segments.
4. Profile each segment's size and value.
5. Recommend an action per segment.
6. Validate that segments are distinct and stable.

Tools & stack

- pandas (qcut/rank)
- seaborn
- a segment summary table

Definition of Done

You are done when all of these are true and demoable

- An RFM segmentation with named segments, sizes, value and recommended actions.
- Demonstrable live on real (even if small) data, not just described.

Pitfalls to avoid

Worry if any of these are true of your work

- Segments with no attached action.
- Equal-width bins on skewed spend.
- Never refreshing, so segments rot.

Where this fits in the track

Over Phase 1 you grow from loading a CSV to shipping a live, decision-grade analytics product. You turn raw data into numbers people trust and decisions people make.